



AlzPredict-Neural Network Approach to Early Diagnosis

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Abstract—*Alzheimer's Disease is a type of dementia that affects thinking, behaviour, and memory. Early diagnosis is important to prevent the disease progression and to get an effective intervention. As we know, Alzheimer's Disease can be treated with MRI scans and cognitive assessments, which will take much time and require an expert for evaluation which can be quite expensive. This project aims to develop a predictive model by using Multilayer Perceptron (MLP) and Back Propagation (BP) to classify the patients based on patients health indicators since it balances simplicity and effectiveness by using machine learning in which the process is much more straightforward than others.*

Keywords — Multilayer Perceptron (MLP) & Back Propagation (BP), Alzheimer's Disease Prediction, Early diagnosis system, Medical Machine Learning

I. INTRODUCTION (*HEADING 1*)

Alzheimer's Disease (AD) is an irreversible, progressive neurodegenerative disorder that predominantly affects the elderly, causing a steady decline in memory, reasoning, and behavior, eventually leading to an inability to perform daily activities. As the most common form of dementia, AD accounts for approximately 60–70% of all dementia cases globally (World Health Organization, 2025). The global impact of dementia is profound, with an estimated

55 million individuals currently living with the condition. This number is projected to rise to 78 million by 2030 and to a staggering 139 million by 2050 (World Health Organization, 2021). According to Alzheimer's Disease International (2024), one person is diagnosed with dementia every three seconds, highlighting the urgency for effective methods of early detection and intervention.

Early diagnosis of AD is essential, as it allows for timely management, increases the quality of life, and provides support for caregivers. However, conventional diagnostic techniques—such as MRI scans, cognitive assessments, and clinical evaluations—are often expensive, time-consuming, and require specialized medical expertise. These limitations are particularly problematic in rural or resource-constrained healthcare settings, where access to diagnostic resources is limited.

To address these challenges, this study proposes the development of an intelligent classification system based on a Multilayer Perceptron (MLP) neural network trained using the Backpropagation (BP) algorithm. This model is designed to classify individuals into high or low risk for developing AD based on multiple features including age, gender, cognitive scores, memory issues, confusion, and family medical history.

The proposed system aims to be scalable, cost-effective, and efficient—providing a valuable tool to support early diagnosis.

II. BACKGROUND

The growing prevalence of Alzheimer's Disease presents a significant public health burden, requiring innovative and accessible approaches to early diagnosis. Traditional diagnostic approaches, while effective, often pose barriers in terms of cost, availability, and complexity. These factors hinder widespread early detection, especially in under-resourced areas (World Health Organization, 2025).

Machine learning, and specifically neural networks, offer promising alternatives in medical diagnostics due to their capacity to detect complex, non-linear relationships within data. The Multilayer Perceptron (MLP), a type of supervised learning model, has been shown to be effective in various medical prediction scenarios. It operates through interconnected layers of neurons and is capable of learning intricate patterns across a variety of input features.

The Backpropagation (BP) algorithm forms the backbone of MLP training, allowing the network to iteratively minimize prediction errors by adjusting weights using gradient descent. Its effectiveness has already been demonstrated in healthcare applications, such as cardiovascular disease prediction (Al-Milli, 2013), showing its ability to handle complex and multifactorial clinical datasets.

This project utilizes the Alzheimer Disease Dataset compiled by El Kharoua (2023) and hosted on Kaggle. This dataset includes over 2,100 records of patient demographic data, medical histories, lifestyle information, and cognitive assessments. By training the MLP on this dataset, the system will classify individuals as either LOW or HIGH RISK of developing AD. The goal is to enhance early-stage decision-making and enable better prioritization of clinical resources.

Although this system is not intended to replace formal medical diagnoses, it offers a valuable decision-support tool for clinicians and healthcare practitioners. Its application can potentially improve diagnostic timelines, expand access to early interventions, and ultimately contribute to better outcomes for patients facing Alzheimer's Disease.

III. METHODOLOGY

The Multilayer Perceptron (MLP) has three primary layers namely the input layer where features are received, hidden layers where processing and extraction of pattern occur and the output layer that outputs the final prediction/classification. The reason that this architecture is best suited to the predictive modeling of Alzheimer's is that

the potential interrelationship between the clinical features of Alzheimer's, including age, blood pressure, memory problems, and other health factors, are non-linear (Goodfellow et al., 2016). Unlike the conventional statistical models, MLPs do not start with any defined assumptions regarding the data distribution and they are able to automatically detect important patterns as the training phase progresses. The notable feature of MLPs is automatic feature extraction. The network can learn the internal parameters (weights and biases) during training to highlight the most relevant features, and thus less manual feature engineering and normalization need to be relied on (LeCun, Bengio, & Hinton, 2015).

In addition, MLP models can handle some level of missing or noisy data, which again makes them applicable in health contexts where data may not be of high quality (Shickel et al., 2018). They can learn with imperfect records, and can control confounding factors implicitly, and thus can encourage strong disease forecasting without detailed person-level evaluations. With the help of these properties, MLP can present a strong framework to identify the risk of Alzheimer disease with multidimensional clinical datasets, which will assist in the early detection of the disease especially in locations where professional examination might not be easily accessible.

Backpropagation is used to train MLP. It works by finding the error between the model's prediction and the actual target through gradient descent. During the model training, the algorithm performs a forward pass first where the input data propagates through a hidden layer to generate an output. Next, the error is calculated backward from the output layer to the input layer for the backward pass and will update the hidden layer weight to reduce prediction error. The gradient-based optimization allows the hidden layer to learn from complex patterns in the data. Backpropagation is crucial for adjusting internal parameters of MLP to enhance its predictive performance. Without backpropagation, MLP cannot effectively learn and predict from the dataset. It already has been proven in medical data to predict heart disease in the study by Al-Milli (2013).

Multilayer Perceptron(MLP) and Back Propagation(BP) to classify the patients based on patients health indicators since it balances simplicity and effectiveness by using machine learning in which the process is much more straightforward than others. The MLP/BP model can accurately differentiate between normal individuals and Alzheimer's Disease individuals. This model can effectively show the individuals with Alzheimer's Disease which will save much time for the doctor to treat the patients earlier than expected.

IV. RESULT AND DISCUSSION

After the training is completed, an assessment was conducted through questionnaires by inputting user-specific

information. The model returned a risk level classification between LOW RISK or HIGH RISK with their confidence scores ranging from 70% to 95%. As an example, user who reported symptoms such as memory complaints, average cognitive scores and forgetful was predicted at HIGH RISK with a confidence score of 91.3% while another user without that symptoms and in a good health was categorized as LOW RISK with a confidence score of 87.6%.

The performance was evaluated using metrics such as accuracy, confusion matrix and precision-recall scores. The performance of the MLP model was evaluated using K-Fold Cross Validation with 5 folds. It is used to assess the generalizability of the MLP across different subsets of data. K-fold ensures that each fold maintains the same proportion of risk levels as in the original dataset which is important to maintain class balance in medical diagnosis tasks.

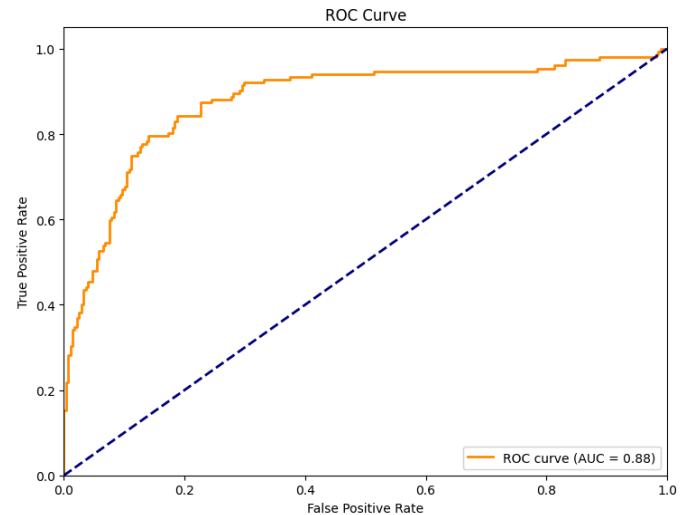


Figure 1.1 ROC Curve

The Receiver Operating Characteristic (ROC) curve visualizes the trade-off between sensitivity (true positive rate) and specificity (true negative rate). The curve moves significantly toward the top-left corner and the Area Under the Curve (AUC) is calculated at 0.88. The AUC indicates that the model has the ability to distinguish between patients with Alzheimer's disease and patients without the disease. AUC values above 0.85 are considered good in medical practice to support the reliability of the MLP model.

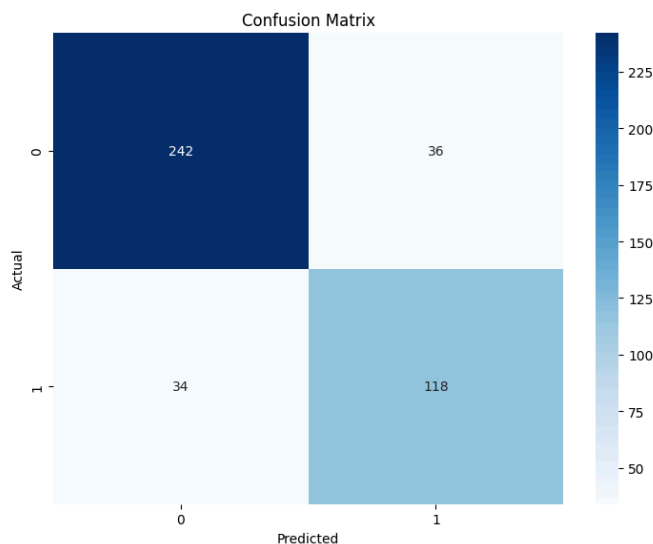


Figure 1.0 Confusion Matrix

The confusion matrix shows that out of 430 samples, the model classified 242 true negatives and 118 true positives correctly while there are 36 false positives and 34 false negatives that are misclassified. The balanced classification performance suggests that the model is essential for medical applications since it is neither overly biased toward high false positives nor missing false negatives.

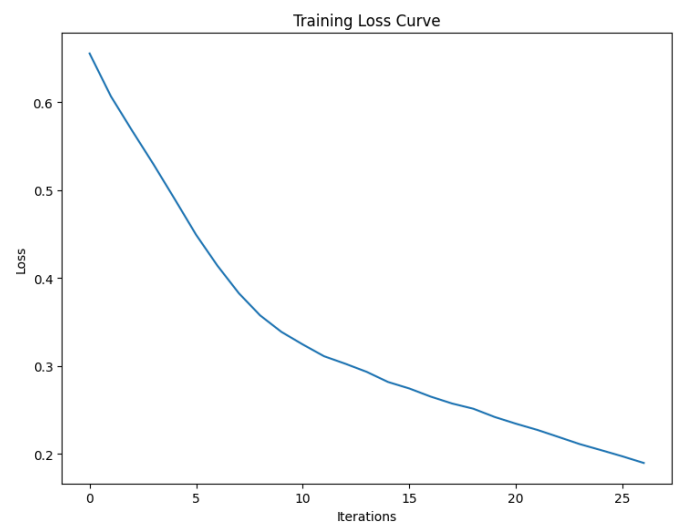


Figure 1.2 Training Loss Curve

Moving onto the training loss curve, it shows a steady decline that indicates an effective learning without overfitting. The prediction probability distribution shows that the model produces confident predictions with most of the outputs skewed toward 0 (LOW RISK) or 1 (HIGH RISK). This

distribution is ideal in binary classification tasks as it shows that the MLP model is not biased toward ambiguous outputs.

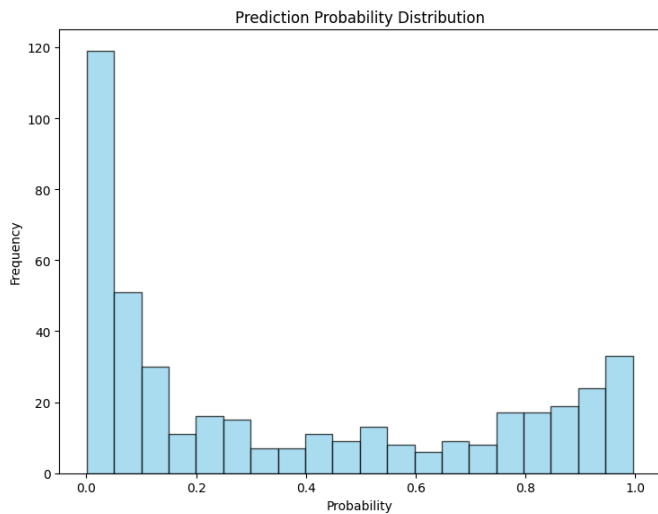


Figure 1.0 Prediction Probability Distribution

Therefore, results prove that the MLP model and Backpropagation are suitable for Alzheimer's disease risk prediction. The MLP model classification performance which is confirmed through multiple validation methods and analysis makes it a good decision support tool for early screening which aligns with prior studies on effectiveness of neural networks in medical diagnosis (Al-Milli, 2013).

V. CONCLUSION

In conclusion, the MLP/BP model effectively handled various health indicators such as age, memory, cognitive test scores and family history. By using supervised learning and gradient-based optimization, the system achieved accurate and reliable classification results thus supporting early detection and risk management for Alzheimer's disease. The use of MLP is highly beneficial due to its ability to learn non-linear relationships from the dataset. The backpropagation mechanism was important in optimizing the MLP prediction performance by adjusting internal parameters to minimize error. The model can be improved by using a larger dataset to enhance the generalizability. It can also be improved by integrating the electronic health records and cognitive test results to improve diagnostic accuracy. Even though this machine learning approach cannot be a replacement for clinical diagnosis, it also can serve as a support system in the early detection of Alzheimer's disease.

ACKNOWLEDGMENT

We consider it our privilege to extend our deepest thanks to our respective lecturer, Dr Alif Ridzuan who has possibly given me constant support, valuable advice and encouragement at all times both during the course or research. Their suggestions and comments proved to be very helpful in determining the way this project went and how successful it was.

We further want to thank El Kharoua (2023) of the publicly available Alzheimer Disease Dataset on Kaggle. Finally, I would like to express gratitude to my family and friends who supported me, drove me and understood me in the process of completing this research.

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